How Much of a Tuned Score Survives on Fresh Items? Optimism and Reliable Reporting for Automatically Tuned Prompts and Pipelines

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Abstract

Automatic prompt and pipeline optimisation reports the benchmark score of the best configuration found, and when that score is measured on the items that chose it, it is optimistic. We measure how far such a score lies above the accuracy of the selected configuration on fresh items, what drives the gap, whether it changes conclusions, and which reporting practice repairs it, by repeating the whole pipeline (draw tuning items, search, select, report) thousands of times against known truth. In a synthetic benchmark with exact truth the naive report is 4.59 pp too high in the reference cell ($n = 200$ tuning items, $K = 16$ candidates of nearly equal quality), 4.8 pp when all candidates are exactly equally good, and 6.7 pp at $K = 128$; the gap is set mainly by n and K and a Gaussian-maximum formula reproduces the simulated gap to within 0.5 pp in 55 cells (with a small systematic underprediction). It changes conclusions: a "tuned beats baseline" claim is licensed in 25% of repetitions in which the tuned configuration is truly no better than the baseline, and systems tuned with unequal budgets are ranked by budget. Holding items out (a split, or tuning on some items and reporting on fresh ones) removes the bias with near-nominal coverage, but at equal total items it costs accuracy: in the reference cell the RMSE is 4.5 pp for 100 tuning plus 100 fresh items, against 2.8 pp for an out-of-bag bootstrap correction on all 200 items (bias -0.1 pp, coverage 98%) and 5.2 pp for the naive report. On a small LLM (Qwen2.5-0.5B-Instruct, MMLU-Pro) with all $K = 60$ configurations of a prompt grid scored (accuracies differing by 1.2 pp) the naive gap is 4.2 pp at $n = 200$ and 9.3 pp at $n = 50$, the Gaussian-maximum formula fed with the real covariance predicts it to about 4%, a synthetic generator matched to the real matrix to within about ten percent, and the protocol ordering at equal items is the synthetic one (RMSE 2.7 pp out-of-bag bootstrap vs. 3.8 pp for $100/100$ tuning/fresh items at 200 items). Successive halving and greedy coordinate search shrink it to 3.0 and 3.4 pp at $n = 200$ but do not remove it. These are replays of searches over a fixed pool of 60, not evidence about LLM-in-the-loop optimisers, larger searches or stronger models. We recommend disclosing the search specification and, where items can be spared, reporting on items not used for selection; where they cannot, an out-of-bag bootstrap correction or a selection-aware interval, whose behaviour we checked in the synthetic setting and on the real model.

1 Introduction

Automatic prompt and pipeline optimisation has become routine: a framework searches over instructions, demonstrations and program structure and returns the configuration that scores best on a set of

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items [Khattab et al., 2023, Opsahl-Ong et al., 2024]. The score of that configuration on the items that chose it is the maximum of many noisy scores and is therefore optimistic in the sense of the winner's curse, more so the more candidates were tried and the fewer items were used [Cawley and Talbot, 2010, Dwork et al., 2015]. Good practice already separates tuning, development and test items, and we do not survey how often it is followed. Our subject is the score on whichever items drove the selection: a validation score that is quoted as the result, or a benchmark that was reused for tuning. For that score we ask how much of it survives on fresh items, which factors drive the difference, whether it changes the conclusions drawn from comparisons, and which reporting practice repairs it. The winner's curse itself is not new. What we add is a measurement, against exact truth and with the whole pipeline repeated, of its size for a discrete (correct/incorrect) outcome with correlated candidates and several search procedures; of its effect on three decisions (a tuned-vs-baseline claim, a comparison of systems tuned with unequal budgets, a ranking of tuned systems); and a side-by-side comparison of reporting methods under one reliability criterion.

We call this difference the *optimism gap* Δ (reported score minus the accuracy of the selected configuration on fresh items) and measure it by repeating the *whole pipeline* thousands of times, so that bias, error and coverage of every reporting method are measured against the truth with Monte-Carlo standard errors. We vary n, K, the true quality spread, candidate correlation, the search procedure, generation noise and leakage of tuning items into the prompt.

Findings (synthetic benchmark).

- In the reference cell ($n = 200$ tuning items, $K = 16$ candidates, true spread $\tau = 0.01$) the naive report is 4.59 ± 0.06 pp above the true accuracy of the selected configuration. The gap exists even when all candidates are exactly equally good ($\tau = 0$: 4.8 pp at $K = 16$) and rises to 6.7 pp at $K = 128$ (reference cell otherwise; Sec. 4.1).
- It is driven mainly by n and K, is reproduced to within 0.5 pp by a Gaussian-maximum formula (slightly underpredicted on average), falls with candidate correlation and with the true quality spread, and is raised by leakage and, mildly, by generation noise.
- It changes conclusions: the naive report licenses "tuned beats baseline" in 25% of repetitions in which the tuned configuration is truly no better than the baseline (boundary of the null: 5%), and systems tuned with unequal budgets are ranked in favour of the larger budget.
- Holding items out gives an unbiased report (the split meets the reliability criterion in 97 of 106 cells), but at equal total items it has a larger error than an out-of-bag bootstrap correction on all items (4.5 vs. 2.8 pp RMSE at 200 items in the reference cell; the bootstrap is better in all 36 combinations we ran, with coverage 91–99%). The simultaneous bound under-covers in only 2 of 106 cells but keeps the naive point bias; cross-fitting under-covers. Our implementations of conditional and hybrid winner-aware inference, validated against a closed form and the authors' code, give valid but wide intervals and a noisy point estimate.

Findings (real LLM). Section 4.2 repeats the measurement on a small open-weight LLM scoring MMLU-Pro under a pool of prompt configurations. All $K = 60$ configurations of a prompt grid are scored (spread 1.2 pp). There the naive report is 4.2 ± 0.1 pp above the held-out accuracy at $n = 200$ and 9.3 ± 0.1 pp at $n = 50$ (the same in an i.i.d. truth arm), the Gaussian-maximum formula fed with the real covariance reproduces these gaps to about 4%, a generator matched to the real matrix reproduces them to within about ten percent, and at equal total items the ordering of the reporting protocols is that of the synthetic study. Successive halving and greedy search shrink it (3.0, 3.4 pp) but do not remove it. These are replays over a fixed pool of 60, not a study of LLM-in-the-loop optimisers or of large K.

Scope. Synthetic statements are about the generator of Sec. 3; real-LLM statements are about one small model, one scoring protocol and one item pool. We did not run DSPy or MIPRO: the searches are our re-implementations of their families, and the conditional and hybrid intervals are our implementations of ideas from Andrews et al. [2024], checked against a port of the authors' replication code but not identical to it.

2 Related work

Selection bias in model selection and test-set reuse. That the best of many noisy estimates is optimistic is classical in machine learning: Cawley and Talbot [2010] show that optimising a model-selection criterion biases the performance estimate reported for the selected model, and argue for nested validation. For adaptive reuse of a holdout, Dwork et al. [2015] give a differential-privacy-based reusable holdout, Blum and Hardt [2015] give the Ladder, a leaderboard mechanism that reports only significant improvements, and Feldman et al. [2019] show that the worst-case bias of k adaptively chosen classifiers shrinks as the number of classes grows, which bears on a 10-option benchmark. Roelofs et al. [2019] measure the gap between a repeatedly queried holdout and a once-used test set empirically on Kaggle competitions. We study the same mechanism for LLM prompt and pipeline search and use these mechanisms as candidate reporting rules or as the comparison for them.

Inference after selection. Andrews et al. [2024] study statistical inference on the selected (winning) estimate; Berk et al. [2013] give post-selection intervals that are valid simultaneously over all possible selections; Chernozhukov et al. [2013] give Gaussian approximations and a multiplier bootstrap for maxima of high-dimensional sums, which gives a simultaneous bound over K configurations. These provide our analytic prediction of the gap and several of our reporting methods; we test them in a setting with discrete outcomes and strongly correlated configurations rather than assume their conditions.

Variability and selection of LLM prompts. Scores of an LLM vary with prompt design: Sclar et al. [2024] report large accuracy spread across semantically equivalent formats and recommend reporting a range, and Wang et al. [2024] report smaller prompt sensitivity on MMLU-Pro than on MMLU. Polo et al. [2024] estimate the *distribution* of performance over many prompt templates from few evaluations, which is a different estimand from the score of the one configuration that is eventually used. Automatic tuning tools choose such a configuration: DSPy [Khattab et al., 2023] compiles prompts and demonstrations for a user-supplied metric, MIPRO [Opsahl-Ong et al., 2024] jointly optimises instructions and demonstrations of multi-stage programs, and Shi et al. [2024] cast prompt selection as fixed-budget best-arm identification. These works study how to *find* a good configuration; we ask what the score of the configuration found on the tuning items says about unseen items, and how to report it.

3 Problem setting and reporting methods

3.1 Setting and quantities

A *system* has a finite set of candidate configurations $\mathcal{C} = \{1, \dots, K\}$ (prompt wordings, demonstration sets, option layouts). A configuration c has an unknown population accuracy μ_c on the benchmark's item distribution. A *tuning set* T of n items is drawn, a *search procedure* $\mathcal{S}$ sees only the correctness of the candidates on T and returns a configuration $\hat{c}$, and the *naive report* is the accuracy of $\hat{c}$ on the same items, $R_{\text{naive}} = \text{acc}_T(\hat{c})$. The target of any honest report is $\mu_{\hat{c}}$, the accuracy of the *selected* configuration on fresh items. We call $\Delta = R_{\text{naive}} - \mu_{\hat{c}}$ the *optimism gap*. It has three sources: selection bias (the maximum of K noisy scores), extra adaptivity of the search (rounds, budget allocation), and leakage when demonstrations are taken from T. We study the first two and, in the synthetic setting, leakage.

The unit of repetition throughout is the *whole pipeline*: draw T, search, select, report, compare with the truth. For a reporting method M with point report $\hat{\theta}_M$ and optional 95% interval, over R repetitions we measure bias $\mathbb{E}[\hat{\theta}_M - \mu_{\hat{c}}]$, RMSE, interval coverage of $\mu_{\hat{c}}$, interval width, and the number of items consumed. Every metric carries a Monte-Carlo standard error, and coverage is compared with the binomial Monte-Carlo band around 0.95, not with 0.95 itself.

3.2 Search procedures

All procedures see an array of $K \times n$ correctness indicators. *Best-of-K* (full evaluation of every candidate on T, argmax with random tie-breaking) models grid or random search. *Successive halving* splits T into $\lceil \log_2 K \rceil$ disjoint blocks and keeps the better half of the survivors each round;

successive rejects is a fixed-budget best-arm identification rule (the framing of prompt selection used by Shi et al. [2024]), run here with a budget of $nK/4$ evaluations; we implement it ourselves and do not run the algorithm of that paper. *Greedy coordinate ascent* over a factorial grid imitates multi-round instruction/demonstration search. In every case the naive report is the winner's score on the evaluations the procedure collected for it.

3.3 Reporting methods

Naive (with Wilson interval). **Held-out split**: select on half of T, report on the other half; unbiased for $\mu_{\hat{c}}$ but it spends half the items and ships a configuration chosen from $n/2$ items. **Fresh set**: select on n_{tune} items and report the winner on m new items. k**-fold cross-fitting** ($k = 5$): run the whole procedure on $k-1$ folds and score its pick on the held-out fold; its estimand is the performance of the *procedure* on $(1-1/k)n$ items. **Bootstrap optimism correction**: resample T, rerun the procedure, and subtract the mean difference between the score on the resample and on T; we also use the out-of-bag and the .632 variants. **Simultaneous bound**: an interval whose critical value is the $(1-\alpha)$ quantile of the bootstrap maximum of $|\hat{p}_k^* - \hat{p}_k|/\mathrm{se}_k$ over all K candidates, valid for any data-dependent selection [Chernozhukov et al., 2013, Berk et al., 2013]; its point report remains the naive score. **Conditional (winner-aware) inference** for the full-evaluation argmax: under a Gaussian approximation of the K scores the selection event is a polyhedron, and inverting the truncated-normal law of the winner's score gives a median-unbiased estimate and an equal-tailed interval, and a hybrid with a projection interval; this follows Andrews et al. [2024] but is our own implementation. **Ladder-type thresholded report** [Blum and Hardt, 2015]: for sequential submission only, the reported best-so-far score moves only when a candidate beats it by more than $1/\sqrt{n}$.

3.4 Gap prediction

For best-of-K with Gaussian tuning scores of covariance Σ/n (the question-level covariance of the candidates' correctness), the expected gap is $\mathbb{E}[\epsilon_{\hat{c}}]$, with $\epsilon \sim N(0, \Sigma/n)$ and $\hat{c} = \arg\max_c(\mu_c + \epsilon_c)$. For independent noise with standard deviation $s = \sqrt{p(1-p)/n}$ and true-accuracy spread τ this is approximately $\frac{s^2}{\sqrt{s^2+\tau^2}}\,\mathbb{E}[\max_{k\le K} Z_k]$ with Z_k standard normal. We use these as empirical approximations to compare with simulation, not as theorems.

3.5 Synthetic and real settings

Synthetic (exact truth). A Gaussian-copula item pool: configuration accuracies $p_c = m + \tau z_c$ with z_c standard normal, an item latent shared across configurations with correlation ρ, and correctness when the latent falls below $\Phi^{-1}(p_c)$. The truth of the selected configuration is its mean over the pool ($N = 20{,}000$ items, 128 configurations; Appendix A). Optional arms add generation noise (each evaluation flipped with probability ε), leakage (a number of tuning items shown with their answers in the prompt) and one outlier configuration.

Real (small LLM, MMLU-Pro). The LLM is the system under tuning: Qwen2.5-0.5B-Instruct (fp32, CPU, no sampling) scores a fixed pool of 2,000 MMLU-Pro test items (revision `b189ec7`, full hash in the reproduction notes) under a pool of prompt configurations that varies the instruction wording, the demonstration set and the option layout (60 configurations), by reading the log-probabilities of the ten option letters at the answer position. Scoring is a single deterministic forward pass, so the correctness matrix is a fixed array and any search can be replayed on it. Tuning sets are drawn from the pool, and the truth is the accuracy of the selected configuration on the pool items not in T (*disjoint* arm; an i.i.d. arm against the pool mean is a check). All real statements are conditional on this pool, model and scoring protocol.

4 Experiments

4.1 Synthetic study with exact truth

Design. The pool has 128 configurations and $N = 20{,}000$ items; the reference cell has $n = 200$ tuning items, $K = 16$ searched configurations, true-accuracy spread $\tau = 0.01$, latent correlation $\rho = 0.5$, mean accuracy 0.3 and best-of-K search. We vary one factor at a time (OFAT, 2000

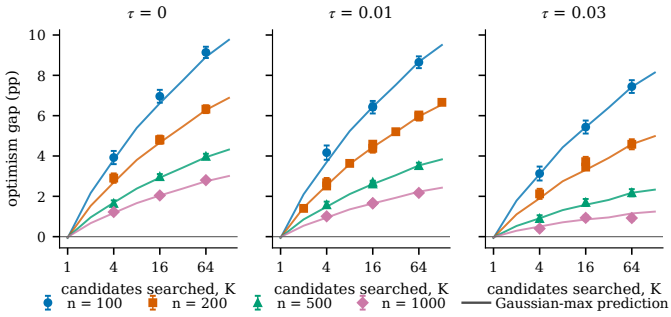


Figure 1: Optimism gap (naive report minus exact true accuracy of the selected configuration, pp) for best-of-K search against the number of candidates K, per tuning-set size n, in three panels by true spread τ ($\rho = 0.5$). Markers: simulation (interaction grid, 500 repetitions per point), bars ± 1.96 Monte-Carlo s.e.; lines: Gaussian-max prediction (Sec. 3). $K = 1$ has no search and no gap. Over the 55 cells with both, observed minus predicted ranges from -0.23 to $+0.47$ pp; the line extends to $K = 128$, where the grid has no simulation.

whole-pipeline repetitions per cell) and run an $n \times K \times \tau \times$procedure interaction grid (500 repetitions per cell) and stress cells (2000), 106 cells in all. The truth of the selected configuration is exact. All gaps are in percentage points (pp) with Monte-Carlo standard errors (s.e.) from the repetitions.

How large is the gap (Q1)? In the reference cell the naive report overstates the fresh-item accuracy of the selected configuration by 4.59 ± 0.06 pp (mean selected true accuracy 30.4%), and the naive 95% Wilson interval covers it in only 76% of repetitions (Table 1, Fig. 7 in Appendix D). The gap exists even when all configurations are equally good ($\tau = 0$: 4.8 pp), so it is a pure selection effect, and it grows with the number of candidates searched ($K = 2$: 1.4 pp; $K = 16$: 4.6 pp; $K = 128$: 6.7 pp) and shrinks with the tuning-set size ($n = 50$: 9.4 pp; $n = 2000$: 1.0 pp; Figs. 1, 2). At the most unfavourable stress cell we ran ($n = 50$, $K = 128$, accuracy near chance) it is 10.4 pp and interval coverage drops to 33%.

What drives it (Q2). Fig. 2 varies each factor from the reference cell. The gap is driven mainly by n and K. A larger true spread τ lowers it ($\tau = 0.03$: 3.4 pp), because a real quality difference makes the selection less noisy. Strong correlation between configurations lowers it ($\rho = 0.9$: 2.8 pp vs. $\rho = 0$: 5.6 pp), since correlated candidates share their noise. Generation noise raises it slightly ($\varepsilon = 0.2$: 5.7 pp) and so does leakage of tuning items into the prompt, which also drives coverage down (8 leaked items: gap 7.0 pp, naive coverage 41%). Among search procedures, successive halving has a smaller gap (3.4 pp) than best-of-K and successive rejects (4.8 pp); it uses part of the data to eliminate and still reports on all of it, but its coverage is still 86%. The Gaussian-max prediction agrees with the simulation in the 55 cells where both exist (mean absolute difference 0.09 pp, at most 0.47 pp; 51 cells have $\rho = 0.5$, and the largest simulated K is 128, in one cell). It slightly *under*predicts: the observed gap is larger in 41 of 55 cells (mean difference $+0.07$ pp; sign test $p \approx 3.6 \times 10^{-4}$), and in the 8 cells that differ by more than two combined s.e. it is larger in 7 and smaller in 1. We use it as an approximation, not a theorem.

Does the gap change conclusions (Q3)? We ask whether a method licenses the claim "the tuned system beats the untuned baseline". The baseline is a fixed pool configuration (index 0, drawn like the others, chosen independently of the candidates but possibly among them by chance), scored on the same tuning items, or on the report items for the held-out split. A method *claims* when (report $-$ baseline score)$/\mathrm{se} > z$, with se the paired standard error of the per-item difference between the selected and the baseline configuration and $z = 1.645$ (one-sided 5%; the simultaneous critical value for the simultaneous bound). The claim is *false* when the selected configuration is in truth not better than the baseline, $\mu_{\hat{c}} \leq \mu_{\text{base}}$, and the false-claim rate is the share of claims among such repetitions (60% of the repetitions in the reference cell). Its null value is the boundary rate 5% at $\mu_{\hat{c}} = \mu_{\text{base}}$; below it the rate should be lower. Requiring a true gain of 0.5 (1) pp for the claim to count as true changes the naive rate in the reference cell only to 26% (26%), and a claim with a negative true difference occurs in 14.8% of all repetitions (split: 2.4%, simultaneous: 0.7%). A *false ship* (Table 1)

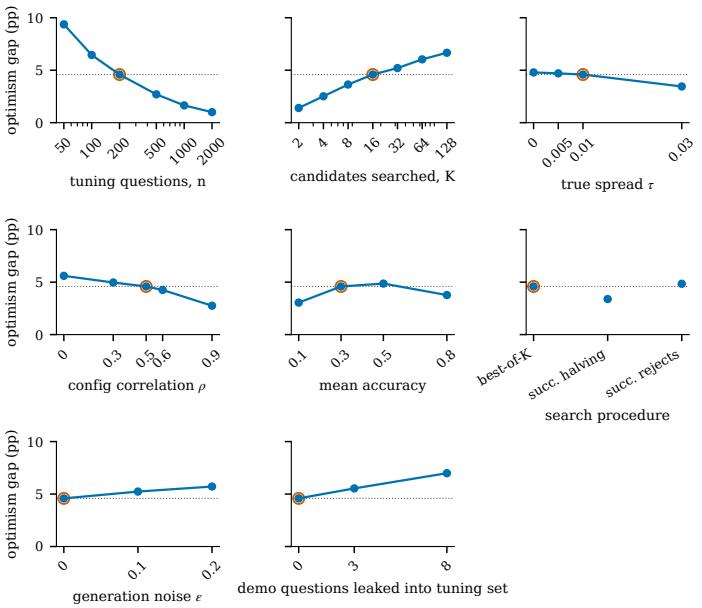


Figure 2: Optimism gap of the naive report when one factor is varied and the others stay at the reference cell ($n = 200$, $K = 16$, $\tau = 0.01$, $\rho = 0.5$, mean accuracy 0.3, best-of-K, no generation noise, no leakage; orange ring and dotted line). 2000 repetitions per point, bars ± 1.96 Monte-Carlo s.e. Greedy search ran only at $K = 128$ and is not comparable.

is a report at least 2 pp above the baseline's true score while the selected configuration's true score is below that bar. In the reference cell the naive report claims in 27% of all repetitions and in 25% of the false cases, against the boundary rate of 5% (Fig. 3a). With $K = 128$ it does in 54%. The comparison of two tuned systems is equally affected when their search budgets differ (Fig. 3b): at $\tau = 0$, i.e. identical systems, a system tuned over 64 configurations is claimed to beat one with a single configuration in 49% of 500 repetitions, and the claim is also not true (true difference not positive) in 20% (46% if a true gain of 0.5 pp is required); the average reported difference is 6.2 pp against a true one of about zero. With equal budgets (16 vs. 16) the false-claim rate is 0.0% at $n = 200$: the bias cancels when both systems were searched alike, and does not when they were not.

Reporting methods (Q4, Q5). Table 1 (Appendix B) and Fig. 7 (Appendix D) give bias, coverage and width in the reference cell. The study plan, written before the runs, fixes a reliability criterion that we apply to all 106 cells and every method: $|\text{bias}| \leq \max(2\,\text{s.e.}, 0.5\text{ pp})$ and 95% coverage inside the binomial Monte-Carlo band around 0.95. The coverage condition is two-sided, so a conservative interval fails it; we therefore also count under-coverage (below the band) and over-coverage (above it). The held-out split meets the criterion in 97 of 106 cells; its 3 bias and 6 coverage failures (2 under, 4 over; largest |bias| 0.64 pp) are of the size expected from testing 106 cells twice at the 5% level. It pays with half of the items (width 17.6 vs. 13.1 pp; RMSE 4.7 vs. 5.3 pp in the reference cell) and reports the configuration chosen from $n/2$ items, so we compare protocols at equal total items below. The naive report meets the criterion in only 1 of 106 cells, under-covering in 100. The simultaneous bound keeps the naive bias (4.57 pp) but under-covers in only 2 cells and over-covers in 102 (98% in the reference cell, width 19.4 pp): valid and conservative. Five-fold cross-fitting under-covers in 105 cells; the bootstrap corrections reduce the bias (out-of-bag: -0.09 pp in the reference cell) but mostly over-cover (67 over, 10 under), so they meet the full criterion in only 26 cells. Our implementations of winner-aware conditional inference and of a hybrid interval (Appendix A) agree with a closed form ($K = 2$) and with a port of the authors' replication code, and are calibrated in exact Gaussian cases (coverage 94.2–96.1%). The conditional interval is nonetheless very wide in the reference cell (width 60.5 pp; RMSE of its median 40.5 pp) because the winner's lead over the truncation boundary is below 0.5 standard errors in 54% of repetitions. The hybrid has coverage 94.0–97.0% at width 17.0 pp (19.7 pp for the simultaneous bound) in the region we ran ($\rho = 0.5$, mean accuracy 0.3, $K \leq 64$, 27 cells), but its point estimate is no better than the naive one. The Ladder-type report

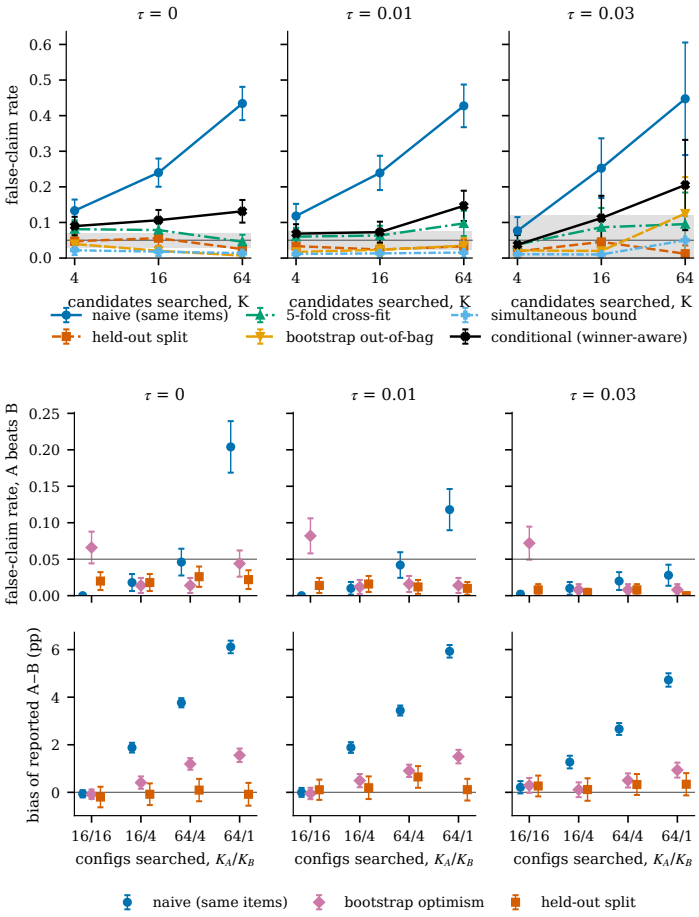


Figure 3: (a) Share of repetitions in which a method licenses "tuned beats untuned baseline" (one-sided 5%) among those where the selected configuration is truly not better ($n = 200$, $\rho = 0.5$, best-of-K; 500 repetitions, bars $\pm 1.96\sqrt{p(1-p)/m}$ with m the repetitions in the denominator; the grey band is the binomial 95% band around the nominal 0.05). (b) Two systems tuned with K_A and K_B configurations, $n = 200$, 500 repetitions. Top: share of repetitions claiming "A beats B" while A is truly not better; line: nominal 0.05. Bottom: bias of the reported $A - B$ difference (pp), bars ± 1.96 s.e.

[Blum and Hardt, 2015] reduces the naive bias but does not remove it ($\tau = 0$, $n = 200$, $K = 16$: 2.4 vs. 4.9 pp).

Reporting protocols at equal item budgets (Q5). We score every protocol against the true accuracy of the configuration *it* ships, at equal total items (Table 2 in Appendix B; 36 combinations of $\tau \in \{0, .01, .03\}$, $K \in \{4, 16, 64\}$, $\rho \in \{.5, .9\}$ and 200 or 400 items, 600 repetitions). A *fresh-set* protocol tunes on n_{tune} items, ships the winner and reports its accuracy on m new items. In the reference cell with 200 items in all, the $100/100$ fresh-set report and the $50/50$ split are unbiased (-0.2 and -0.0 pp) with 95% coverage, but their RMSE is 4.5 and 4.6 pp because the report items carry their own sampling error; the best fresh allocation ($50/150$) has RMSE 3.9 pp; the out-of-bag bootstrap on all 200 items has bias -0.1 pp, RMSE 2.8 pp and coverage 98%, the naive report $+4.5$ pp and 5.2 pp. The out-of-bag bootstrap has the smaller RMSE than the split and than the half-and-half fresh set in all 36 combinations, with coverage 91–99% (conservative in most), while the hybrid's RMSE exceeds the naive one in 31. Holding items out thus buys an unbiased, assumption-free report, not a more accurate one. The out-of-bag bootstrap needs the per-item outcomes of every searched candidate and a search that can be replayed on resampled items; we checked it only for the searches here.

Ranking several tuned systems. For five systems tuned on the same items with 4, 8, 16, 32 and 64 configurations at $\tau = 0$ (all equally good, $n = 200$, 500 repetitions; Fig. 6 in Appendix D), the naive report puts the largest-budget system first in 39% of repetitions, whereas it is truly best in 15% (the held-out split: 12% vs. 13%); with five systems tuned alike (16 configurations each) the naive rate is 30%. When quality differs ($\tau = 0.03$) the naive rank correlation with the truth is not worse than the split's (Kendall's τ_b 0.49 vs. 0.42); the distortion is that the budget, not the quality, orders equally good systems. At an equal total budget of 400 items the out-of-bag bootstrap, the split and the half-and-half fresh set put the 64-candidate system first in 13, 12 and 13% of repetitions at $\tau = 0$ (36% for the naive report; truly best in 15%), so they remove the budget effect, but with true differences ($\tau = 0.03$) none ranks better than the naive report by Kendall's τ_b (0.50, 0.52, 0.51 vs. 0.60; 500 repetitions).

4.2 Real LLM on MMLU-Pro

Scale reached. The pool is all $K = 60$ configurations of a factorial grid of instruction wording, demonstration set and option layout, each scored on all 2000 items by Qwen2.5-0.5B-Instruct. Their accuracies are 15.2–20.8% (mean 18.2%; chance 10%), a spread of 1.2 pp (s.d.), with a mean item-level correlation of 0.35 between configurations. This is a small-spread regime with one small model; it is not a study of large K, of stronger models or of DSPy or MIPRO, which we did not run. Searches are replayed on the cached $0/1$ matrix (1000 repetitions per cell). In the *disjoint* arm ($n \leq 400$) the tuning items are drawn without replacement and the truth is the selected configuration's accuracy on the remaining $2000 - n$ items; in the *i.i.d.* arm they are drawn with replacement and the truth is the accuracy over the whole pool ($n \leq 800$).

The gap is present and has the same shape. With exhaustive search over the $K = 60$ configurations the naive report exceeds the held-out accuracy of the selected configuration by 9.3 ± 0.1 pp at $n = 50$, 6.4 at $n = 100$, 4.2 ± 0.1 at $n = 200$ and 2.8 pp at $n = 400$, and the i.i.d. arm gives the same (9.0, 4.1, 2.6 pp; Fig. 4). At $n = 200$ it grows with the number of candidates ($K = 1$: -0.0 ± 0.1, $K = 2$: 1.1, $K = 4$: 2.1, $K = 60$: 4.2 pp), although the search has little to find (mean regret 2.0 pp). At $n = 200$ the cheaper searches shrink the gap but do not remove it: successive halving 3.0 pp, greedy coordinate search over the factorial grid 3.4 pp, successive rejects 3.6 pp, against 4.2 pp for exhaustive search. The naive Wilson interval covers the truth in only 71% of repetitions at $n = 200$ (64% at $n = 50$), and the naive report licenses "tuned beats the baseline" in 25% of the repetitions at $n = 200$ in which the selected configuration is truly no better (27% at $n = 50$; nominal 5%).

The Gaussian-maximum formula and the generator, checked on the real matrix. Fed with the empirical covariance of the real matrix, the formula of Sec. 3 predicts the replayed gap at $K = 60$ as 8.9 vs. 9.3 pp ($n = 50$), 4.2 vs. 4.2 (200) and 2.7 vs. 2.8 pp (400; Fig. 4a); over all 63 cells ($K \leq 60$, n up to 800, both arms) the difference is at most 0.36 pp, with 6 cells beyond two Monte-Carlo s.e. Generator settings matched to the real accuracies, mean item correlation (latent $\rho = 0.58$) or the full pairwise correlations reproduce the real naive gap to within about ten percent (one-ρ copula +7% on average, pairwise copula -1% for $K \geq 4$), not exactly (Appendix E).

Reporting methods. Table 3 (Appendix B) gives the real-setting version of the reference table. At $n = 200$ the held-out split is unbiased (-0.1 ± 0.1 pp, coverage 94%), the out-of-bag bootstrap reduces the bias to -0.3 pp at nearly the naive width, the hybrid interval covers 97% with RMSE 5.2 pp (naive 4.8) and the conditional interval, 56.2 pp wide, covers 95%. At equal total items (200; Table 4) the RMSE is 2.7 pp for the out-of-bag bootstrap (coverage 94%), 3.4 and 3.8 pp for 50/150 and 100/100 tuning/fresh items, 3.9 pp for the split and 5.1 pp for the naive report: the order of the synthetic study.

5 Discussion and limitations

What to report. The measurements support three practices for any result obtained after an automatic search. (i) *Disclose the search*: the number of candidates, the number of tuning items, the procedure and its seeds; the synthetic gap is a function of n and K first (Figs. 1, 2), and a tuned-vs-untuned or tuned-vs-tuned comparison is not like-for-like unless the budgets are given (Fig. 3). (ii) *Report on items not used for selection when they can be spared*: select on part of the items and report on a

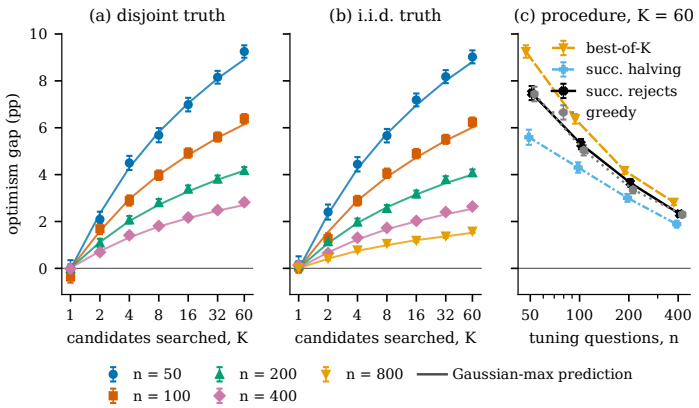


Figure 4: Real setting (Qwen2.5-0.5B-Instruct, MMLU-Pro, $K \leq 60$ complete configurations, 1000 repetitions per point, bars ± 1.96 Monte-Carlo s.e.). Optimism gap of the naive report of best-of-K search against the number of candidates searched, one colour per tuning-set size n; markers are the replay, lines the Gaussian-maximum prediction from the empirical covariance of the real matrix; $K = 1$ is the no-search control. (a) Disjoint truth arm. (b) I.i.d. truth arm (truth over the whole pool). (c) At $K = 60$, best-of-K, successive halving, successive rejects and greedy coordinate search against n (disjoint arm).

separate set. This is unbiased and needs no modelling of the search. At equal total items it is less accurate than correcting the in-sample score (RMSE 4.5 pp for 100 tuning plus 100 fresh items vs. 2.8 pp for the out-of-bag bootstrap at 200 items, Table 2), so its price is accuracy, not only width. (iii) *When items cannot be spared*, either correct the score or use an interval that accounts for the selection. The out-of-bag bootstrap had the smallest error and a small bias in the synthetic reference cell and across the 36 equal-budget combinations (coverage 91–99%, mostly conservative), and had bias -0.3 ± 0.1 pp, coverage 97% and, at 200 items, the smallest RMSE (2.7 vs. 3.8 pp for 100/100 tuning/fresh items) on the real model ($K = 60$); it requires the per-item outcomes of every candidate searched and a search that can be replayed on resampled items, which an adaptive optimiser may not allow. The hybrid winner-aware interval was near-nominal at a width near that of the simultaneous bound in the region where we ran it, but its point estimate was no better than the naive one (RMSE 5.6 vs. 5.2 pp at 200 items), and the conditional interval was too wide to inform at the sample sizes studied.

Limitations. *Synthetic setting.* The generator is a Gaussian copula with a finite configuration pool; its conclusions hold for that generator and the ranges of n, K, τ, ρ and mean accuracy we scanned (Sec. 4.1), and the Gaussian-maximum prediction is an approximation that deviates by up to 0.5 pp. On the real covariance it deviates by at most 0.36 pp, and a matched generator reproduces the real gap to about ten percent, not exactly (Sec. 4.2). *Real setting.* The pool is $K = 60$ configurations with a spread of 1.2 pp, a factorial grid rather than a random sample of prompts, and the searches (best-of-K, successive halving, successive rejects, greedy coordinate search) are replayed over that fixed pool rather than run with the LLM in the loop; a real leakage arm, a real two-system comparison and a per-subject analysis were not run, and there is no check at large K and no stronger model. One small model with letter-logit scoring; the truth is the held-out remainder of a finite pool. We did not run DSPy or MIPRO and say nothing about their search dynamics. *Methods.* The conditional and hybrid intervals are our implementations; they pass the checks of Appendix A (closed form, exact-Gaussian coverage, a port of the authors' replication code) but their coverage in the data-generating setting was run only at $\rho = 0.5$, mean accuracy 0.3 and $K \leq 64$ for the full-evaluation argmax. The equal-budget comparison covers 200 and 400 items and best-of-K search; the out-of-bag bootstrap was not run on adaptive search. Cross-fitting estimates the performance of the procedure run on a fraction of the data, not of the final configuration. False-claim and ranking rates depend on the claim threshold and the share of no-better configurations (varied only through τ). Optimistic tuned scores can lead to deploying prompts worse than reported; correct reporting reduces that risk, and we see no direct negative use.

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A Exact definitions of generators, procedures and estimators

These definitions follow the code in `analysis/` (`synth.py`, `lib.py`, `run_grid.py`); the table of scripts is in `REPRODUCE.md`. Throughout, $S \in \{0,1\}^{K\times n}$ holds the correctness of the K candidates on the n tuning items, $\hat{p}_c$ is the row mean, $\hat{c}$ the selected row, and ties in an argmax are broken at random.

Synthetic generator. A pool of 128 configurations and $N = 20{,}000$ items. Configuration accuracies are $p_c = \mathrm{clip}(m + \tau z_c, 0.01, 0.99)$ with $z_c \sim N(0,1)$ ($m = 0.3$ unless stated); an optional outlier raises one configuration's accuracy by a fixed amount. Each item has a latent $q_i \sim N(0,1)$ shared across configurations, and configuration c has the latent $U_{ci} = \sqrt{\rho}\, q_i + \sqrt{1-\rho}\, e_{ci}$ with $e_{ci} \sim N(0,1)$; it is correct on item i iff $U_{ci} < \Phi^{-1}(p_c)$, so every row has accuracy p_c in expectation and any two rows have latent correlation ρ. The true accuracy μ_c of a configuration is its mean over the N pool items (the pool is the population). A tuning set is n items drawn with replacement; the K candidates are drawn without replacement from the 128 configurations. *Generation noise* flips each evaluation of a candidate (and of the baseline) with probability ε independently, and the truth becomes $\mu_c + \varepsilon(1 - 2\mu_c)$. *Leakage*: each candidate shows ℓ tuning items with their answers in the prompt, which turns an incorrect outcome on such an item into a correct one with probability 0.9; the truth is unchanged, and the held-out split scores its report half without leakage.

Search procedures. *Best-of-K*: $\hat{c} = \arg\max_c \hat{p}_c$ on all n items. *Successive halving*: $R = \lceil \log_2 K \rceil$ rounds; the n items are cut into R consecutive blocks; in round r the survivors are scored on block r only and the better $\lceil |\text{alive}|/2 \rceil$ stay; the report is the survivor's score on all n items. *Successive rejects*: budget $B = nK/4$ evaluations, $\bar{\ell} = \frac{1}{2} + \sum_{i=2}^{K} 1/i$; in phase $j = 1, \ldots, K-1$ the survivors are scored on the first $n_j = \min\{n, \max\{1, \lceil (B-K)/(\bar{\ell}(K+1-j)) \rceil\}\}$ items and the worst is removed; the report is the winner's mean over the first n_{K-1} items. *Greedy coordinate ascent*: the K candidates form a factorial grid ($4 \times 4 \times 4$, or $8 \times 4 \times 4$ for $K = 128$); start at a random cell and, for the factors in random order, move to the best level given the others on all n items, for at most 10 sweeps; the report is the final cell's score.

Claim test and baseline. The baseline is pool configuration 0, scored on the same n items as the candidates (on the report items for the held-out split), with the same generation noise. For any method with point report $\hat{\theta}$ and critical value z (1.645; the simultaneous critical value c for the simultaneous bound), the claim is $(\hat{\theta} - \hat{p}_{\text{base}})/\mathrm{dse} > z$, where dse is the standard deviation (ddof 1) of the per-item differences $S_{\hat{c},i} - S_{\text{base},i}$ divided by $\sqrt{n_{\text{rep}}}$ (n_{rep} the number of report items). The false-claim rate is the mean of the claim indicator over repetitions with $\mu_{\hat{c}} \le \mu_{\text{base}}$. A false ship is $\hat{\theta} \ge \mu_{\text{base}} + 0.02$ and $\mu_{\hat{c}} < \mu_{\text{base}} + 0.02$.

Reporting methods. All intervals are 95% ($\alpha = 0.05$) and Wilson intervals use the number of items the report was computed from. *Naive*: $\hat{p}_{\hat{c}}$ with a Wilson interval on n. *Held-out split*: the search runs on the first $\lfloor n/2 \rfloor$ items (the tuning set is in random order), the report is the pick's mean on the remaining items with a Wilson interval on their number. *Five-fold cross-fitting*: n items in five consecutive folds; for each fold the procedure runs on the other four and its pick is scored on the fold; the report is the pooled score over all n items, with a Wilson interval on n; the configuration it describes is the full-data pick. *Bootstrap*: $B = 100$ resamples of the n items with replacement; for each, the procedure is run on the resample, giving pick $\hat{c}_b$; the optimism is the score of $\hat{c}_b$ on the resample minus its score on the original items, and the optimism-corrected report is the naive score minus the mean optimism; the out-of-bag report is the mean score of $\hat{c}_b$ on the items absent from resample b; the .632 report is $0.368\,\hat{p}_{\hat{c}} + 0.632\,\mathrm{OOB}$; intervals are the report $\pm 1.96\sqrt{\hat{\theta}(1-\hat{\theta})/n}$, i.e. they ignore the selection. *Simultaneous bound*: $B = 300$ bootstrap resamples of the items; c is the 0.95 quantile of $\max_k |\hat{p}^*_k - \hat{p}_k|/\mathrm{se}_k$ with $\mathrm{se}_k = \sqrt{\max(\hat{p}_k(1-\hat{p}_k), 10^{-4})/n}$; the interval is $\hat{p}_{\hat{c}} \pm c\,\mathrm{se}_{\hat{c}}$ and the point report is naive. *Ladder*: $\eta = 1/\sqrt{n}$; candidates are visited in a random order; the first sets the ladder value $\mathrm{round}(\hat{p}/\eta)\eta$, a later one replaces the selected configuration and the value only if $\hat{p} >$ value $+ \eta$; the report is the final value with a Wilson interval on n.

Conditional and hybrid intervals. Both treat $\hat{p} \sim N(\mu, \hat{\Sigma}/n)$ with $\hat{\Sigma}$ the sample covariance of the rows of S, and apply to best-of-K only. To remove lattice ties each score is dithered by one

uniform draw on $\pm 1/(2n)$ per repetition, used for every quantity of that repetition, which leaves the argmax a random-tie argmax. Given the winner $\hat{c} = k$, the event $\{\hat{c} = k\}$ is a polyhedron in $\hat{p}_k$ once the component of $\hat{p}$ orthogonal to $\hat{p}_k$ is fixed (the polyhedral lemma of Lee et al. [2016], in the form for the winner of a maximum [Andrews et al., 2024]); its truncation interval $[V^-, V^+]$ is computed from the $K - 1$ contrasts. *Conditional*: the conditional law of $\hat{p}_k$ is $N(\theta, \sigma_k^2)$ truncated to $[V^-, V^+]$; the estimate is the θ at which the conditional CDF at the observed value equals $1/2$ and the interval endpoints are the θ at which it equals $1 - \alpha/2$ and $\alpha/2$ (root-finding on $\hat{p}_k \pm 80\sigma_k$). *Hybrid*: $\beta = 0.005$; c_β is the $1 - \beta$ quantile (from 1000 Monte-Carlo draws) of $\max_j |Z_j|$ with $Z \sim N(0, \mathrm{corr}(\hat{\Sigma}))$; for each θ_0 on a grid of 161 values in $\hat{p}_k \pm c_\beta\sigma_k$ the truncation interval is intersected with $[\theta_0 - c_\beta\sigma_k, \theta_0 + c_\beta\sigma_k]$, and θ_0 is retained if the conditional CDF of $\hat{p}_k$ lies in $[a_1, 1 - a_1]$ with $a_1 = (\alpha - \beta)/(2(1 - \beta))$; the interval is the range of retained values and the point estimate is the median of the truncated normal on the restricted set (the hybrid median-unbiased estimate, always within $\hat{p}_k \pm c_\beta\sigma_k$). The conditional point estimate is the unrestricted conditional median. *Validation.* (1) For $K = 2$ the conditional CDF equals an independent closed form (computed with mpmath) to $1e - 15$ over 2000 draws. (2) With exact Gaussian draws and known Σ (6 cases, $K = 2$–16, τ from 0 to .05, n from 200 to 1000, 3000 repetitions) the probability-integral transform of the conditional CDF at the true score is uniform (Kolmogorov–Smirnov p-values 0.27–0.88) and both intervals cover 94.2–96.1%. (3) A Python port of the authors' MATLAB replication code for the conditional and hybrid routines agrees on 361 random covariance matrices (39 more were skipped because the port's truncated-normal quantile saturates): truncation limits to 10^{-12}, conditional limits and median to about 10^{-6} standard errors, hybrid limits to within one step of our 161-point grid (the port uses 4001 points) and the hybrid median to 0.04 of a step. (4) In the reference cell with known Σ the conditional interval covers 96% (with Σ estimated from Bernoulli draws: 95%), with mean width 60 and median width 33 pp.

Real replay. The correctness matrix $Y \in \{0, 1\}^{K \times 2000}$ is fixed. Each repetition draws n items without replacement as the tuning set (candidates are all K complete configurations, or a random subset), runs a procedure and a reporting method on them, and takes as truth the selected configuration's accuracy on the other $2000 - n$ items. The baseline for claims is the hand-written default (instruction 0, demonstration set 1, plain layout; accuracy 19.15%), which is one of the candidates when all K are searched. Greedy coordinate search is run on the $6 \times 5 \times 2$ grid of instruction, demonstration set and layout (all 60 configurations), with the same random start and sweeps as in the synthetic study. In the i.i.d. arm the n items are drawn with replacement and the truth is the accuracy of the selected configuration on all 2000 items. The Gaussian-maximum prediction for the real matrix uses $\epsilon \sim \mathcal{N}(0, \hat{\Sigma}/n)$ with $\hat{\Sigma}$ the empirical covariance of the 0/1 item matrix of the K candidates (5000 draws). In the disjoint arm the truth is the accuracy on the $N - n$ remaining items ($N = 2000$), so the tuning noise has covariance $(1 - n/N)\hat{\Sigma}/n$ and the predicted gap is $N/(N - n)$ times the expected noise of the selected configuration. The generator matched to the real matrix (Appendix E) uses the real accuracies as thresholds of a Gaussian copula with either one latent correlation or the pairwise tetrachoric correlation matrix of the real 0/1 data (eigenvalues clipped to be positive semi-definite)..

B Further reporting-method results

The other methods mostly fail the criterion, for different reasons. The naive report fails in all but 1 of 106 cells, under-covering in 100. The simultaneous bound keeps the naive bias (4.57 pp) but under-covers in only 2 cells and over-covers in 102 (98% in the reference cell), at a width of 19.4 pp: it is valid and conservative. Five-fold cross-fitting has bias near zero in 103 cells but under-covers in 105 (87% in the reference cell) because its interval ignores the variance added by selection. The bootstrap optimism correction reduces the bias from 4.59 to 1.27 pp and its out-of-bag variant to -0.09 pp (bias criterion met in 94 cells) but over-covers in 67 cells (97% in the reference cell) and under-covers in 10, so it meets the full criterion in only 26 of 106. Our implementation of winner-aware conditional inference was run in 67 cells; it covers correctly in the reference cell (95%) and is outside the band in 3 (under) and 7 (over) of them, but its intervals are very wide (60.5 pp), its heavy-tailed conditional median has RMSE 40.5 pp and it meets the bias criterion in 0 of them. The instability is not a coding fault (Appendix A): the conditional median is median-unbiased, but when the winner's standardised lead over the truncation boundary is small the truncated normal is nearly flat over a long left tail in the score, so the mean error is large. In the reference cell with known Σ the lead is below 0.5 standard errors in 54% of repetitions (median 0.45); in the 32% with a lead below

0.25 the conditional median has median error -15 pp and mean error -38 pp and its interval has median width 105 pp, while the hybrid has mean error -7 pp at width 14 pp. Over all repetitions the point-estimate RMSE is 43 pp for the conditional and 5.3 pp for the hybrid median (median absolute error 4.5 and 4.1 pp). With one clearly better configuration ($+5$ pp, $K = 16$, $n = 500$) the winner is the truly best in 85% of repetitions (98% at $n = 1000$, where the conditional interval is 8 pp wide, as wide as the naive one). The conditional interval is thus a correct but uninformative answer when the data barely separate the winner from the rest. A hybrid interval of the same family (projection interval plus the conditional test on the restricted truncation set) has coverage 94.0–97.0% at a mean width of 17.0 pp over 27 cells of the $\tau \times n \times K$ grid (500 repetitions each; Fig. 5 in Appendix D), against 48.4–94.8% at 13.3 pp for the naive interval and 97.0–99.6% at 19.7 pp for the simultaneous bound. It needs the full-evaluation argmax and a Gaussian approximation and was checked only at $\rho = 0.5$, mean accuracy 0.3 and $K \leq 64$; we do not recommend it for its point estimate. The Ladder-type report [Blum and Hardt, 2015] (sequential submission) reduces the naive bias but does not remove it ($\tau = 0$, $n = 200$, $K = 16$: 2.4 vs. 4.9 pp).

Table 1: Reporting methods in the reference cell, exact truth. Every row is scored against the true accuracy of the configuration it reports on: the selected configuration for all rows except the held-out split, which reports on the configuration chosen from $n/2$ items (Table 2 compares protocols at equal total items).

method	bias (pp)	RMSE (pp)	coverage	width (pp)	false claim	false ship
naive (same items)	4.59 ± 0.06	5.25	0.759 ± 0.010	13.1	0.248	0.818
held-out split	0.07 ± 0.11	4.72	0.946 ± 0.005	17.6	0.036	0.327
5-fold cross-fit	0.00 ± 0.09	4.23	0.866 ± 0.008	12.6	0.070	0.278
bootstrap optimism	1.27 ± 0.06	3.14	0.966 ± 0.004	13.2	0.051	0.356
bootstrap out-of-bag	-0.09 ± 0.06	2.86	0.974 ± 0.004	12.7	0.017	0.207
bootstrap .632	1.63 ± 0.06	3.18	0.960 ± 0.004	12.9	0.058	0.411
simultaneous bound	4.57 ± 0.06	5.25	0.983 ± 0.003	19.4	0.012	0.818
conditional (winner-aware)	-10.45 ± 0.87	40.45	0.952 ± 0.005	60.5	0.091	0.354

Reference cell: n=200, K=16, τ=.01, ρ=.5, mean accuracy .3, best-of-K; 2000 repetitions. Bias and coverage are mean $\pm$ Monte Carlo s.e.; coverage is of the nominal 95% interval for the true score of the selected configuration; false claim: rate of claiming tuned $>$ baseline when it is not truly better; false ship: rate of reporting a score at least 2 pp above the baseline's true score while the selected configuration's true score is below that bar.

C Additional synthetic results

D Additional figures

Figures 5–7 support the reporting-methods paragraph and the ranking paragraph of Sec. 4.1; their numbers are in the result files named in `REPRODUCE.md`.

E Generator calibrated to the real matrix

To check that the synthetic generator is informative about real tuned scores, we built generator settings that match the real matrix and compared their naive gap with the real replay (Table 8). Both settings use the 60 real accuracies as the configuration accuracies and as the thresholds of a Gaussian copula over items: (i) one common latent correlation, chosen ($\rho = 0.58$) to reproduce the mean pairwise φ of the real 0/1 matrix; (ii) the full matrix of pairwise tetrachoric correlations of the real data (19 eigenvalues clipped to make it positive semi-definite). Each generator is run on 10 pools of 2000 items with 300 repetitions each, with the replay procedure of the real study. Mean accuracy, the spread of accuracies, the dispersion of item difficulty (standard deviation across items of the share of configurations solving the item: 0.232 real, 0.231 generator) and the mean φ are matched by the one-ρ copula (the pairwise copula shares the thresholds). The one-ρ copula cannot reproduce the heterogeneity of the pairwise φ (real -0.01–0.93; largest pairwise error 0.57) and overstates the gap at $K \geq 4$; the pairwise copula matches every pair (largest error 0.053) and understates it slightly, probably through higher-order structure that a Gaussian copula does not carry. At mean accuracy 0.195, latent $\rho = 0.3, 0.5, 0.7, 0.9$ correspond to an observed φ of $0.16, 0.29, 0.45, 0.69$; the real $\varphi = 0.35$ lies between the grid values $\rho = 0.5$ and 0.9 of the synthetic study, and the grid's mean

Table 2: Reporting protocols at equal total items, each scored against its own shipped configuration (reference cell).

protocol	items	true acc. (%)	bias (pp)	RMSE (pp)	coverage	width (pp)
naive	200	30.44	4.53 ± 0.10	5.19	0.762 ± 0.017	13.1
bootstrap out-of-bag	200	30.46	-0.12 ± 0.12	2.85	0.975 ± 0.006	12.7
bootstrap optimism	200	30.46	1.23 ± 0.12	3.13	0.958 ± 0.008	13.2
simultaneous bound	200	30.45	4.51 ± 0.10	5.18	0.970 ± 0.007	19.4
5-fold cross-fit	200	30.48	-0.04 ± 0.18	4.42	0.848 ± 0.015	12.6
hybrid (proj.+cond.)	200	30.44	-0.88 ± 0.23	5.63	0.940 ± 0.010	16.5
held-out split 50/50	200	30.29	-0.02 ± 0.19	4.57	0.952 ± 0.009	17.6
fresh set 50/150	200	30.20	-0.04 ± 0.16	3.87	0.940 ± 0.010	14.5
fresh set 100/100	200	30.34	-0.23 ± 0.18	4.53	0.952 ± 0.009	17.6
naive	400	30.59	3.16 ± 0.08	3.66	0.788 ± 0.017	9.2
bootstrap out-of-bag	400	30.57	-0.06 ± 0.08	2.07	0.973 ± 0.007	9.0
bootstrap optimism	400	30.57	0.91 ± 0.08	2.26	0.962 ± 0.008	9.3
simultaneous bound	400	30.57	3.18 ± 0.07	3.67	0.978 ± 0.006	13.7
5-fold cross-fit	400	30.57	0.14 ± 0.12	2.98	0.862 ± 0.014	9.0
hybrid (proj.+cond.)	400	30.58	-0.36 ± 0.16	4.00	0.950 ± 0.009	11.5
held-out split 50/50	400	30.42	0.17 ± 0.14	3.34	0.928 ± 0.011	12.6
fresh set 100/300	400	30.30	0.16 ± 0.10	2.54	0.950 ± 0.009	10.3
fresh set 200/200	400	30.46	0.26 ± 0.12	2.99	0.960 ± 0.008	12.7

Reference cell (τ=.01, K=16, ρ=.5, mean accuracy .3, best-of-K), 600 repetitions; the budget (items) is the total number of questions consumed: methods above the fresh-set rows tune and report on the same items (in-sample), a fresh-set row "n_{tune}/m" tunes on n_{tune} items and reports the winner on m new items (Wilson interval). Each row is scored against its own truth, the true accuracy of the configuration that protocol shipped (third column; it differs across protocols because the search saw different data). Bias and coverage are mean $\pm$ Monte Carlo s.e.; coverage is of the nominal 95% interval; the held-out split tunes on one half of the items and reports on the other.

accuracy of 0.3 instead of 0.18 changes φ at equal ρ by about 0.03 (0.32 vs. 0.29 at $\rho = 0.5$). Over all 63 cells ($K = 1, \ldots, 60$, n up to 800, both arms) the one-ρ copula differs from the replay by more than two Monte-Carlo s.e. in 31 cells and the pairwise copula in 4.

F Real-LLM setup and compute record

Table 9 records the model, data revision, scoring protocol and compute of the real-LLM study as read from the committed compute log.

NeurIPS Paper Checklist

1. **Claims**

   Question: Do the main claims made in the abstract and introduction accurately reflect the paper's contributions and scope?

   Answer: [Yes]

   Justification: Abstract and introduction state the synthetic findings with their ranges, and say that the real-LLM part is a replay of searches over $K = 60$ configurations of one small model and that DSPy/MIPRO were not run (Secs. 1, 4.1, 4.2).

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   - The answer [N/A] means that the abstract and introduction do not include the claims made in the paper.
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   - The claims made should match theoretical and experimental results, and reflect how much the results can be expected to generalize to other settings.
   - It is fine to include aspirational goals as motivation as long as it is clear that these goals are not attained by the paper.

Table 3: Reporting methods in the real setting (best-of-K with $K = 60$ complete configurations, disjoint truth).

n	method	bias (pp)	RMSE (pp)	coverage	width (pp)
50	naive (same items)	9.25 ± 0.13	10.19	0.641 ± 0.015	24.0
	held-out split	-0.15 ± 0.24	7.57	0.960 ± 0.006	28.6
	5-fold cross-fit	0.17 ± 0.21	6.73	0.896 ± 0.010	20.8
	bootstrap optimism	2.56 ± 0.15	5.40	0.985 ± 0.004	24.7
	bootstrap out-of-bag	-0.17 ± 0.14	4.50	0.978 ± 0.005	21.2
	bootstrap .632	3.30 ± 0.14	5.48	0.978 ± 0.005	22.7
	simultaneous bound	9.27 ± 0.13	10.20	0.995 ± 0.002	38.6
	conditional (winner-aware)	-25.69 ± 2.64	87.26	0.964 ± 0.006	130.4
	hybrid (proj.+cond.)	-1.51 ± 0.32	10.20	0.971 ± 0.005	32.4
200	naive (same items)	4.18 ± 0.07	4.80	0.709 ± 0.014	11.6
	held-out split	-0.07 ± 0.13	4.00	0.944 ± 0.007	15.0
	5-fold cross-fit	-0.09 ± 0.12	3.76	0.843 ± 0.012	10.7
	bootstrap optimism	0.97 ± 0.08	2.80	0.964 ± 0.006	11.6
	bootstrap out-of-bag	-0.32 ± 0.08	2.61	0.966 ± 0.006	10.7
	bootstrap .632	1.34 ± 0.08	2.83	0.952 ± 0.007	11.1
	simultaneous bound	4.17 ± 0.08	4.80	0.988 ± 0.003	18.4
	conditional (winner-aware)	-11.04 ± 1.23	40.50	0.953 ± 0.007	56.2
	hybrid (proj.+cond.)	-0.85 ± 0.16	5.19	0.966 ± 0.006	15.3

Real-LLM replay on MMLU-Pro with Qwen2.5-0.5B-Instruct: best-of-K over the K=60 configurations, tuning set of n questions drawn without replacement from the 2000-question pool; 1000 repetitions per row. Truth = accuracy of the selected configuration on the pool questions outside the tuning set (finite-pool truth). Bias and coverage are mean $\pm$ Monte Carlo s.e.; coverage is of the nominal 95% interval. The bias and RMSE of the conditional method are dominated by the heavy tail of its estimate (reported as measured).

Table 4: Reporting protocols at equal total items on the real matrix, each scored against its own shipped configuration.

protocol	items	true acc. (%)	bias (pp)	RMSE (pp)	coverage	width (pp)
naive	200	18.82	4.46 ± 0.11	5.10	0.650 ± 0.021	11.6
bootstrap out-of-bag	200	18.81	-0.05 ± 0.12	2.74	0.944 ± 0.010	10.8
bootstrap optimism	200	18.81	1.23 ± 0.12	3.00	0.956 ± 0.009	11.7
simultaneous bound	200	18.83	4.45 ± 0.11	5.09	0.986 ± 0.005	18.4
5-fold cross-fit	200	18.81	0.08 ± 0.16	3.65	0.856 ± 0.016	10.7
hybrid (proj.+cond.)	200	18.82	-0.46 ± 0.22	4.99	0.966 ± 0.008	15.4
held-out split 50/50	200	18.54	0.08 ± 0.18	3.91	0.948 ± 0.010	15.0
fresh set 50/150	200	18.46	-0.04 ± 0.15	3.36	0.936 ± 0.011	12.3
fresh set 100/100	200	18.58	0.05 ± 0.17	3.76	0.950 ± 0.010	15.0

Real-LLM replay on MMLU-Pro (Qwen2.5-0.5B-Instruct), best-of-K with K=60 configurations (largest K in the file), total budget 200 questions, 500 repetitions. Questions are drawn without replacement from the 2000-question pool; each row is scored against the accuracy of the configuration it shipped on the pool questions that protocol did not use (third column, true accuracy of the shipped configuration). Columns as in the synthetic table.

2. **Limitations**

Question: Does the paper discuss the limitations of the work performed by the authors?

Answer: [Yes]

Justification: Section 5 lists the limits (synthetic generator, one small model and a finite pool, the reduced K, our own implementations of conditional/hybrid inference, validation ranges of the hybrid interval).

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- The paper should point out any strong assumptions and how robust the results are to violations of these assumptions (e.g., independence assumptions, noiseless settings, model well-specification, asymptotic approximations only holding locally). The authors

Table 5: Naive optimism gap by tuning-set size and number of candidates.

τ	n	K=4	K=16	K=64
0	100	3.92 ± 0.17	6.96 ± 0.16	9.14 ± 0.14
0	200	2.90 ± 0.12	4.81 ± 0.10	6.32 ± 0.10
0	500	1.65 ± 0.08	2.97 ± 0.07	3.98 ± 0.07
0	1000	1.22 ± 0.05	2.04 ± 0.05	2.80 ± 0.05
0.03	100	3.13 ± 0.18	5.44 ± 0.16	7.44 ± 0.16
0.03	200	2.12 ± 0.13	3.72 ± 0.12	4.58 ± 0.13
0.03	500	0.90 ± 0.08	1.70 ± 0.09	2.18 ± 0.09
0.03	1000	0.40 ± 0.06	0.93 ± 0.06	0.92 ± 0.06

Naive optimism gap (pp; reported minus true score of the selected configuration), mean $\pm$ Monte Carlo s.e. over 500 repetitions; ρ=.5, best-of-K.

Table 6: Stress cells.

cell	method	bias (pp)	coverage	width (pp)	false claim
n=50, K=128, acc. 0.1	naive (same items)	10.41 ± 0.08	0.330 ± 0.011	21.9	0.550
	held-out split	0.15 ± 0.13	0.967 ± 0.004	23.7	0.056
	5-fold cross-fit	0.07 ± 0.13	0.908 ± 0.006	16.5	0.060
	bootstrap out-of-bag	0.04 ± 0.07	0.990 ± 0.002	16.5	0.004
	simultaneous bound	10.40 ± 0.08	0.999 ± 0.001	41.9	0.000
	conditional (winner-aware)	-28.54 ± 1.83	0.965 ± 0.004	138.5	0.117
n=200, K=16, outl. 0.03	naive (same items)	4.74 ± 0.05	0.757 ± 0.010	13.1	0.257
	held-out split	-0.09 ± 0.10	0.949 ± 0.005	17.6	0.042
	5-fold cross-fit	0.08 ± 0.09	0.867 ± 0.008	12.6	0.085
	bootstrap out-of-bag	0.01 ± 0.06	0.975 ± 0.003	12.7	0.025
	simultaneous bound	4.74 ± 0.06	0.973 ± 0.004	19.4	0.015
	conditional (winner-aware)	-10.23 ± 0.83	0.950 ± 0.005	62.3	0.104
n=200, K=128, outl. 0.03	naive (same items)	6.88 ± 0.05	0.409 ± 0.011	13.3	0.535
	held-out split	-0.04 ± 0.10	0.953 ± 0.005	17.6	0.034
	5-fold cross-fit	-0.02 ± 0.09	0.875 ± 0.007	12.6	0.082
	bootstrap out-of-bag	-0.13 ± 0.06	0.988 ± 0.002	12.7	0.016
	simultaneous bound	6.87 ± 0.05	0.986 ± 0.003	23.5	0.014
	conditional (winner-aware)	-13.07 ± 0.97	0.957 ± 0.005	77.5	0.150
n=200, K=16, outl. 0.05	naive (same items)	4.59 ± 0.06	0.767 ± 0.009	13.1	0.252
	held-out split	-0.01 ± 0.10	0.955 ± 0.005	17.6	0.042
	5-fold cross-fit	0.02 ± 0.09	0.875 ± 0.007	12.6	0.082
	bootstrap out-of-bag	-0.14 ± 0.06	0.975 ± 0.003	12.7	0.022
	simultaneous bound	4.58 ± 0.06	0.983 ± 0.003	19.4	0.015
	conditional (winner-aware)	-9.78 ± 0.82	0.966 ± 0.004	61.0	0.094
n=200, K=128, outl. 0.05	naive (same items)	6.10 ± 0.07	0.505 ± 0.011	13.3	0.546
	held-out split	0.08 ± 0.10	0.951 ± 0.005	17.7	0.041
	5-fold cross-fit	-0.31 ± 0.10	0.846 ± 0.008	12.7	0.082
	bootstrap out-of-bag	-0.76 ± 0.07	0.939 ± 0.005	12.8	0.017
	simultaneous bound	6.08 ± 0.07	0.981 ± 0.003	23.6	0.014
	conditional (winner-aware)	-13.77 ± 1.04	0.953 ± 0.005	72.8	0.152

Stress cells (τ=0, ρ=.5, best-of-K; outlier = fraction of outlier configurations); 2000 repetitions each. Columns as in the reference-cell table (RMSE and false-ship rate omitted for width; they are in the results file); bias and coverage are mean $\pm$ Monte Carlo s.e.; outl. = outlier-configuration fraction, acc. = mean accuracy (.3 unless shown).

should reflect on how these assumptions might be violated in practice and what the implications would be.

- The authors should reflect on the scope of the claims made, e.g., if the approach was only tested on a few datasets or with a few runs. In general, empirical results often depend on implicit assumptions, which should be articulated.
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Table 7: Two tuned systems with unequal search budgets.

τ	K_A/K_B	claim (naive)	false claim	wrong sign	reported diff. (pp)	true diff. (pp)	false claim (split)
0	16/16	0.000 ± 0.000	0.000	0.000	-0.05	0.01	0.020
0	16/4	0.034 ± 0.008	0.018	0.018	1.84	-0.04	0.018
0	64/4	0.114 ± 0.014	0.046	0.046	3.77	0.01	0.026
0	64/1	0.492 ± 0.022	0.204	0.202	6.17	0.06	0.022
0.01	16/16	0.000 ± 0.000	0.000	0.000	-0.02	-0.01	0.014
0.01	16/4	0.038 ± 0.009	0.010	0.010	2.16	0.28	0.016
0.01	64/4	0.122 ± 0.015	0.042	0.042	3.82	0.38	0.012
0.01	64/1	0.532 ± 0.022	0.118	0.118	6.74	0.81	0.010
0.03	16/16	0.016 ± 0.006	0.002	0.002	0.33	0.12	0.008
0.03	16/4	0.130 ± 0.015	0.010	0.010	2.75	1.48	0.004
0.03	64/4	0.318 ± 0.021	0.020	0.020	5.16	2.49	0.008
0.03	64/1	0.736 ± 0.020	0.028	0.028	8.99	4.27	0.000

Two tuned systems A and B searched with K_A and K_B configurations (best-of-K, n=200); 500 repetitions per row. Naive columns: claim = rate of claiming A>B (one-sided 5%), false claim = claim while A is truly not better (true difference $\leq$ 0), wrong sign = claim while A is truly worse; mean reported and true difference A$-$B in pp. Last column: false-claim rate of the held-out split.

Table 8: Generator calibrated to the real matrix against the real replay.

quantity	real matrix	one-rho copula	pairwise copula
accuracy, mean (%)	18.23	18.21	(same thresholds)
accuracy, sd across configs (pp)	1.16	1.17	(same thresholds)
item difficulty, sd across items	0.232	0.231	–
mean pairwise ϕ	0.348	0.344	–
pairwise ϕ, min–max	-0.007–0.926	0.321–0.366	–
pairwise ϕ, max $\lvert\Delta\rvert$ over pairs	–	0.572	0.053
naive gap, n=200, K=2 (pp)	1.09 ± 0.09	1.14 ± 0.05 (+0.5)	1.09 ± 0.05 (-0.0)
naive gap, n=200, K=4 (pp)	2.07 ± 0.08	2.04 ± 0.05 (-0.4)	1.97 ± 0.05 (-1.1)
naive gap, n=200, K=8 (pp)	2.81 ± 0.08	2.95 ± 0.05 (+1.5)	2.67 ± 0.05 (-1.6)
naive gap, n=200, K=16 (pp)	3.39 ± 0.08	3.57 ± 0.04 (+2.1)	3.31 ± 0.05 (-0.9)
naive gap, n=200, K=32 (pp)	3.81 ± 0.08	4.30 ± 0.04 (+5.6)	3.83 ± 0.05 (+0.2)
naive gap, n=200, K=60 (pp)	4.18 ± 0.07	4.85 ± 0.04 (+7.8)	4.14 ± 0.04 (-0.4)
generator $-$ real, mean ($K \geq 4$, %)	–	$+6.8$	-1.3
generator $-$ real, range ($K \geq 4$, %)	–	-10 to $+16$	-14 to $+4$
cells with $\lvert z\rvert > 2$	–	31 of 63	4 of 63

Gaussian-copula generator calibrated to the real matrix (accuracies as thresholds): one common latent correlation ("one-rho", matched to the mean pairwise ϕ) or one per pair ("pairwise copula", tetrachoric correlations, 19 eigenvalues clipped). Gap rows: disjoint truth arm, mean $\pm$ s.e. (pp) of the generator (10 pools $\times$ 300 replications) with z against the real replay in parentheses. Summary rows pool all n (50, 100, 200, 400, 800) and both truth arms (disjoint and iid). The copula cannot reproduce the heterogeneity of the real pairwise dependence (near-duplicate and near-independent pairs) and, for the pairwise version, higher-order structure.

used reliably to provide closed captions for online lectures because it fails to handle technical jargon.

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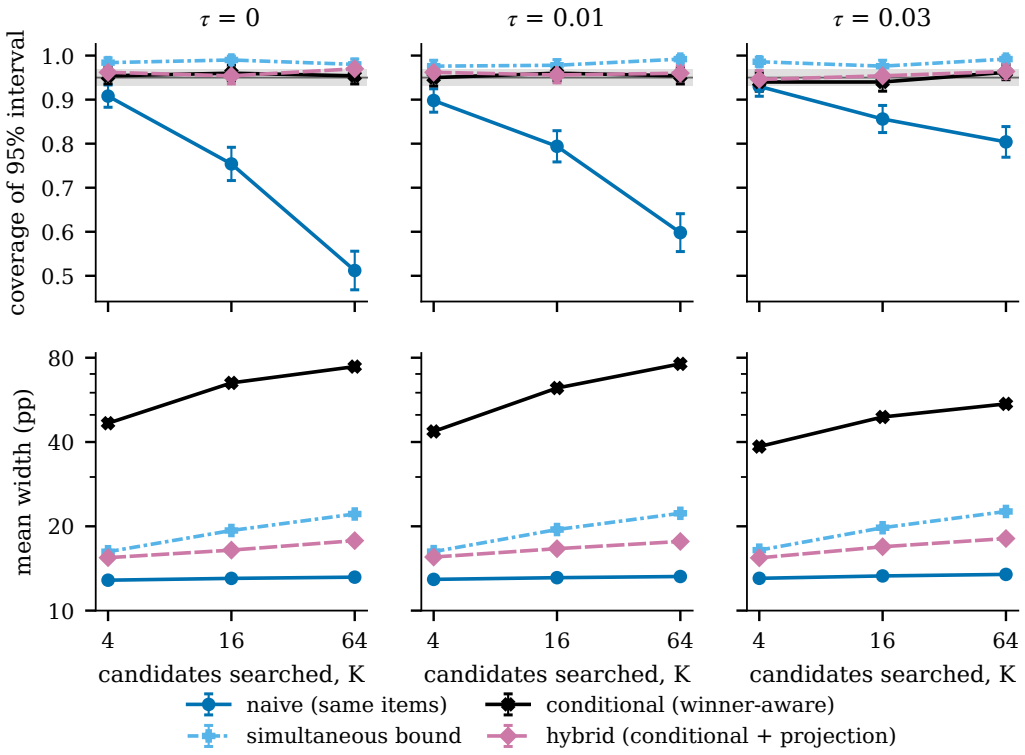


Figure 5: Winner-aware intervals for the argmax winner ($n = 200$, $\rho = 0.5$, 500 repetitions). Coverage of the nominal 95% interval for the true score of the selected configuration (bars ± 1.96 s.e.; grey band: binomial Monte-Carlo band) and mean width (log scale) for the naive, simultaneous, conditional and hybrid intervals.

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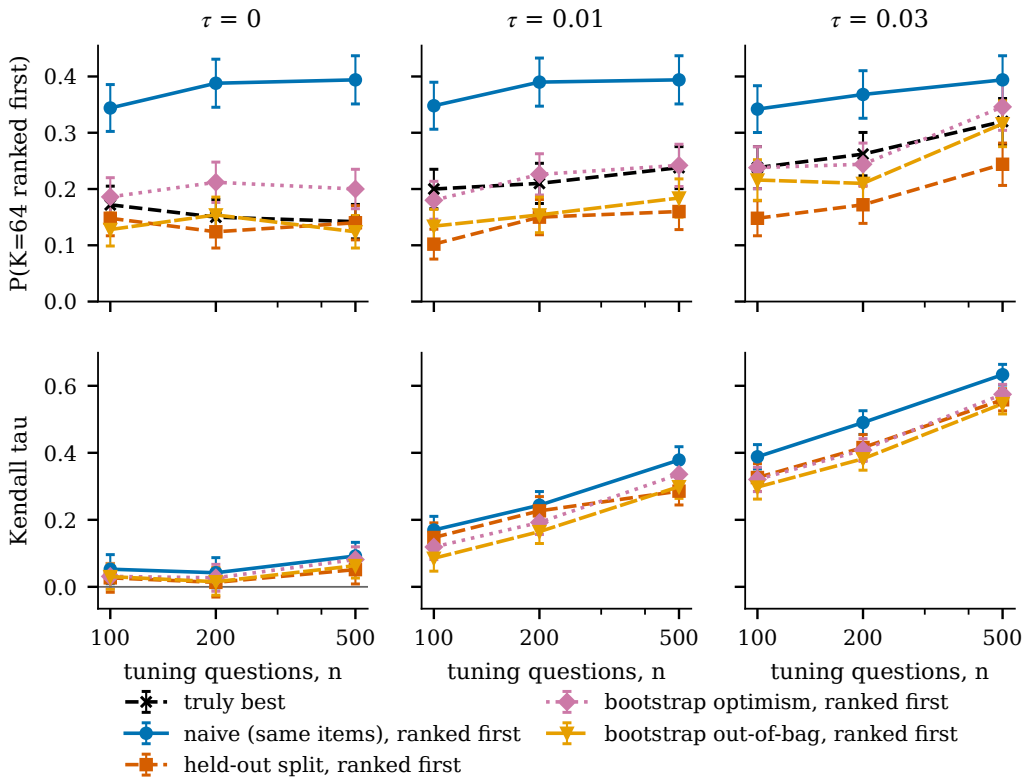


Figure 6: Ranking five systems tuned on the same items with $K = 4, 8, 16, 32, 64$ configurations (best-of-K, 500 repetitions). Top: share of repetitions in which each method ranks the $K = 64$ system first, against how often it is truly best (black dashed). Bottom: mean Kendall's τ between reported and true ranking.

Justification: Generators, pool construction, procedures, reporting methods and settings are specified in Sec. 3; seeds, versions, commands and the figure/table-to-script map are in the reproduction notes (`REPRODUCE.md`) shipped with the code.

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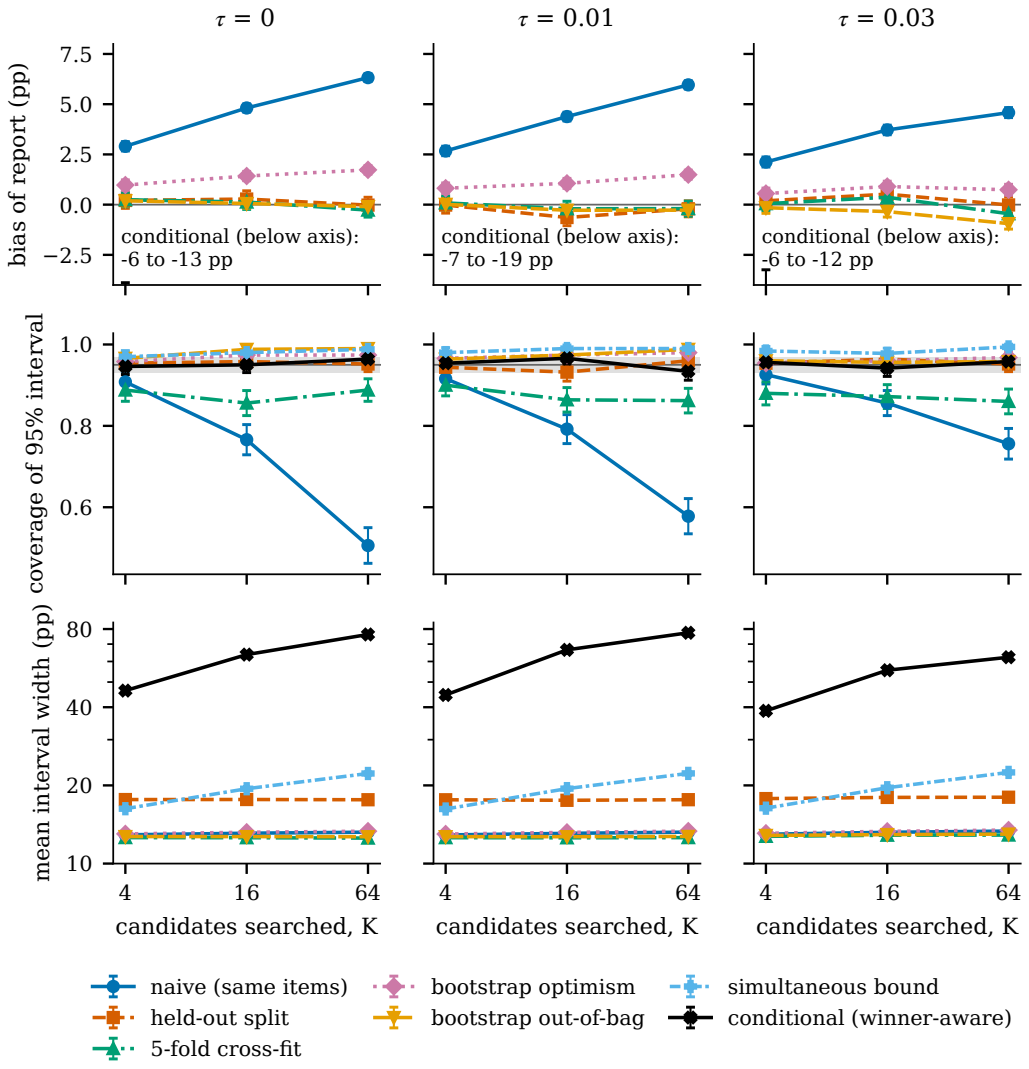


Figure 7: Reporting methods against the exact truth ($n = 200$, $\rho = 0.5$, best-of-K, 500 repetitions). Top: bias of the reported score. Middle: coverage of the nominal 95% interval (grey band: binomial 95% Monte-Carlo band). Bottom: mean interval width (log scale). The conditional method lies below the plotted bias range (see panel text) and has intervals 40–80 pp wide.

Table 9: Setup and compute record of the real-LLM study (read from the compute log).

item	value
model	Qwen2.5-0.5B-Instruct, fp32, CPU; option-letter log-probabilities
benchmark	MMLU-Pro test, revision `b189ec765aa7`, 2000 questions (seed 2024)
configurations	60 complete of 60 planned (instruction wording $\times$ demo set $\times$ layout)
forward passes	120,000 scored of 120,000 planned
accuracy across configs	mean 18.2%, range 15.2–20.8%, sd 1.16 pp
item correlation	mean pairwise correlation of item correctness across configs 0.35
scoring jobs	10 scoring jobs of 6 configurations each, all finished; wall-clock 20.7–34.7 h per job (0.7 h for the job resumed after an out-of-memory stop)
replay jobs	real replay K=60 disjoint (argmax/sh/sr/greedy), M=1000 85 min; real replay K=60 iid, M=1000 109 min
seeds	configs/demos 12345, questions 2024, replay 5
packages	Python 3.11.2, numpy 2.4.6, scipy 1.17.1, pandas 3.0.6, scikit-learn 1.9.1, torch 2.14.1+cpu, transformers 5.22.1

Setup and compute record of the real-LLM matrix, from the compute log and the matrix summary; all 60 planned configurations are scored. Scoring wall-clock is elapsed time in the job on a shared host and includes time the container was frozen, so it exceeds the CPU time used. Replay times are the logged running times.

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   Answer: [Yes]

   Justification: MMLU-Pro is public at a pinned revision and the code, cached correctness matrix and result files ship with the submission; the notes give the exact commands (a public link is withheld for anonymity).

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   Answer: [Yes]

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